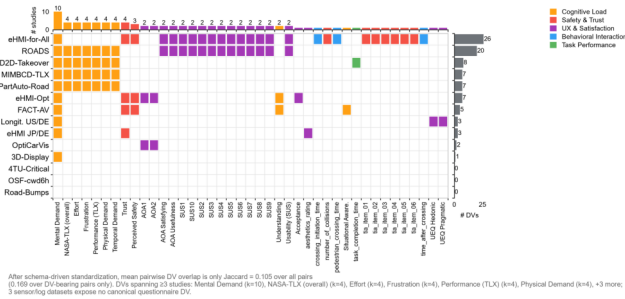


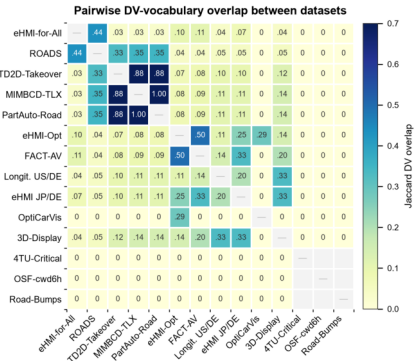
OpenDV-HCI: standardizing dependent variables turns fragmented open datasets into comparable, synthesizable evidence

(a) DV coverage across 14 datasets

Dependent-variable coverage across 14 open HCI datasets



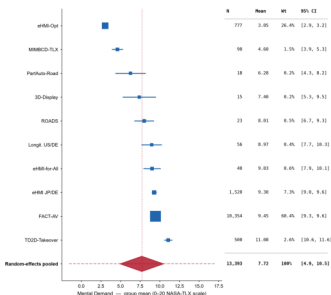
(b) Pairwise overlap (Jaccard)



(c) Meta-analysis: Mental Demand

Evidence synthesis unlocked by standardization: NASA-TLX Mental Demand

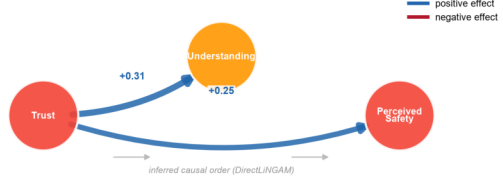
Pooling is only possible because heterogeneous source labels were mapped to one canonical DV. In a participant-level random-effects meta-analysis, high-Q of diverse dataset sources can still reach statistical significance differences.



Random effects (DerSimonian-Laird), k = 14 studies, heterogeneity: $I^2 = 96.4\%$, $\tau^2 = 14.47$, $Q(13) = 639.0$, $p < 0.001$

(d) Cross-study causal structure

Cross-study causal structure among standardized DVs



DirectLINGAM on pooled within-study z-scores (complete-case, non-synthetic; n = 10,354 obs). Edge weights are structural coefficients.